

transparent material. The waves basically squeeze and stretch the material, changing its index of refraction.

Holographic technology projects 3D light images, and these projections may be used to display 3D images. The science of creating holograms is known as holography, and the secret behind this futuristic technology is light. The man who discovered the principles supporting holography was physicist and Nobel Prize (1971) winner Dennis Gabor. Dennis Gabor conducted several holography experiments that he called wave-front reconstruction. Early days of holographic technology may have been limited as Gabor resorted to using electrons to create imagery. However, modern holography seems to benefit from the use of lasers, which may have propelled Gabor's underlying approach into a new era of innovation.

The early prototypes of tele-immersive displays require users to wear special goggles and a head device that tracks the viewpoints of users looking at the screen. On the other end, the people that appear as 3D images are tracked with an array of seven ordinary video cameras, while two other video cameras capture real light patterns projected in each room to calculate distances. This enables the proper depth to be re-created on the screen. So, if a viewer moves his head to the right, he can see the corresponding images that would be seen if he were actually in the room with the person on the screen.

Images on the screen are split and polarised to create a different image for each eye. The goggles then combine these images so that the brain recognises only one 3D image. If that rate could be improved to 10 frames per second, it would create a seamless projected image that would be like looking through a window at another person. Scientists are developing new technologies (including the Internet, display technologies, haptic sensors and desktop supercomputers) to support this type of communication.

Electronics with holography

Since its inception in the 1940s, holography seems to be gradually making its way toward integration with eve-



Holographic display (Credit: <http://bigthink.com>)

ryday devices and may potentially become a part of our daily lives. The potential applications of 3D digital holograms are enormous. In addition to arts and entertainment, various fields including biomedical imaging, scientific visualisation, engineering design and displays could benefit from this technology. For example, creating full-sized organs for 3D analysis by doctors could be helpful, but it remains a challenge owing to the limitation of hologram-generation techniques.

Researchers are looking to create flexible and elastic thin films that could be used on a whole range of surfaces, opening up the horizons of holographic applications. This achievement in holographic technology may grant electronic devices with advanced optical images and potentially many other uses.

Tele-immersion will blur the lines between real and computer-generated images. It will be the ultimate tele-commuting technology, almost entirely eliminating the rush-hour drive to work. Instead of commuting, people could attend board meetings by projecting themselves into the company's conference room.

Holography promises to change how we approach technology. From medical diagnostics to education, data storage, defence and cyber security, 3D holography has the potential to transform a range of industries. There is anticipated interest from consumers who may wish to watch

their favourite movies from their smartphones, projected off screen in 3D. Several tech companies are evaluating smartphones and wearables in preparation for integration of holographic sensors.

Some current applications

1. Holographic interferometry is used by researchers and industry designers to test and design many things, from tyres and engines to prosthetic limbs and artificial bones and joints.

2. Supermarket and department store scanners use a holographic lens system that directs laser light onto the bar codes of the merchandise.

3. Holographic optical elements are used for navigation by airplane pilots. A holographic image of the cockpit instruments appears to float in front of the windshield. This allows the pilot to keep his eyes on the runway or the sky while reading the instruments. This feature is available on some models of automobiles.

4. Medical doctors can use three-dimensional holographic CAT scans to make measurements without invasive surgery. This technique is also used in medical education.

5. Holograms are used in advertisements and consumer packaging of products to attract potential buyers.

6. Holograms have been used on covers of magazine publications, as well as sports trading cards.

7. Use of holograms on credit cards and debit cards provides added security to minimise counterfeiting.